Pre-training Distillation for Large Language Models: A Design Space Exploration

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Abstract

Knowledge distillation (KD) aims to transfer knowledge from a large teacher model to a smaller student model. Previous work applying KD in the field of large language models (LLMs) typically focused on the post-training phase, where the student LLM learns directly from instructions and corresponding responses generated by the teacher model. In this paper, we extend KD to the pre-training phase of LLMs, named **pre-training distillation** (PD). We first conduct a preliminary experiment using GLM-4-9B as the teacher LLM to distill a 1.9B parameter student LLM, validating the effectiveness of PD. Considering the key impact factors of distillation, we systematically explore the design space of pre-training distillation across four aspects: logits processing, loss selection, scaling law, and *offline* or *online* logits. We conduct extensive experiments to explore the design space of pre-training distillation and find better configurations and interesting conclusions, such as larger student LLMs generally benefiting more from pre-training distillation, while a larger teacher LLM does not necessarily guarantee better results. We hope our exploration of the design space will inform future practices in pre-training distillation.

1 Introduction

Knowledge distillation (KD; Hinton, 2015) aims to distill the knowledge of a large teacher model into a smaller and efficient student model for model compression (Gou et al., 2021). It has been widely applied in computer vision (Ahn et al., 2019; Tian et al., 2020; Bergmann et al., 2020; Zhao et al., 2022), natural language processing (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020a; Xu et al., 2024), and speech recognition (Chebotar and Waters, 2016; Fukuda et al., 2017; Tan and Wang, 2021) domains. In recent years, knowledge distillation has been a standard practice to enhance large language models (LLMs) with knowledge from more advanced LLMs, such as GPT-4 (OpenAI, 2023). This technique is typically used during the post-training stage of LLMs, where the student model learns directly using language modeling (LM) loss from a set of queries and responses generated by teacher LLMs. Post-training KD is simple and widely applicable, leading to the development of various advanced LLMs (Taori et al., 2023; Vicuna, 2023; Sun et al., 2024; Cui et al., 2024), which significantly advances the development of LLMs. The success of post-training distillation raises the question of whether distillation LLMs in the pre-training stage is feasible.

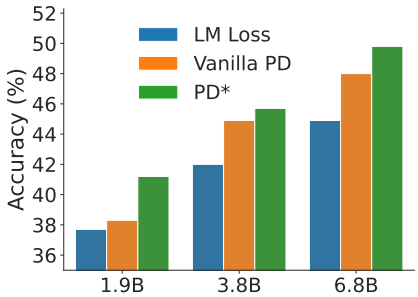

Figure 1: Results of the pre-trained 1.9B, 3.8B, and 6.8B student LLMs, using only LM loss, vanilla PD configuration (§ 3.1), and a better PD configuration (PD*) after our exploration. Details are placed in appendix A.6.

In this paper, we extend knowledge distillation to the pre-training phase of LLMs, named pre-training distillation (PD). We primarily investigate pre-training with **logits-based** KD (Gou et al., 2021), where the student model learns from the teacher model generated logits of each token in the pre-training corpora using a KD loss, such as Kullback–Leibler divergence. The intuition is that the logits from the teacher model contain richer information and can serve as label smoothing (Gou et al., 2021), which could potentially accelerate the training of the student LLM and enhance its performance. Although the potential advantage of

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pre-training distillation is clear, there is limited exploration on how to better apply PD. Therefore, in this paper, we take an initial step in exploring the design space of pre-training distillation. Considering the key factors impacting distillation, we explore the design space of PD in four aspects: (1) **Logits processing**, focusing on the post-processing of the teacher LLM's logits to reduce the memory overhead, including truncation and normalization. (2) **Loss selection**, focusing on the choice of pre-training distillation loss. (3) **Scaling law**, covering varying sizes of student and teacher LLMs, as well as pre-training corpus size. (4) ***Offline* or *online***, meaning logits are generated either from a pre-trained teacher LLM (*offline*) or simultaneously during the pre-training of teacher LLM (*online*). Figure 1 illustrates the effectiveness of the explored better PD configuration (PD*).

We conduct extensive experiments to explore the design space of PD. Specifically, we first conduct a preliminary study using GLM-4-9B (GLM et al., 2024) as the teacher model to generate logits for 100 billion tokens, distilling a 1.9B student LLM from scratch using negative log-likelihood loss. Due to the large vocabulary size (about 150k) of GLM-4-9B, we truncate the logits using top-p-k truncation to reduce storage space: first using top-p (Holtzman et al., 2019) truncation with $p = 0.95$, followed by top-100 truncation. The truncation reduces storage space by $4,000\times$ to about 15 TB of disk space. The preliminary PD yields an average performance improvement of 1.6% across a comprehensive set of English and Chinese datasets, compared to standard pre-training with LM loss, which demonstrates the effectiveness of PD. Based on the preliminary experiment, we explore the design space of PD using controlled experiments: (1) **Logits processing**. We investigate the impact of different p and k values on top-p-k truncation results, and different normalization temperatures. We find no significant difference between various p and k values, with smaller p or k effectively reducing logits storage. The temperature for normalization should not be too high, and adaptive temperature shows no significant benefit. (2) **Loss selection**. We explore the choice of KD loss and the combination of KD loss with LM loss. We find that Kullback–Leibler divergence and negative log-likelihood loss result in similar improvements, but MSE loss suffers a significant drop. The best combination of LM and KD loss is using the Warmup-Stable-Decay (WSD; Hu et al., 2024) method to schedule for the proportion of KD loss, paired with a WSD learning rate scheduler. This suggests that using a higher proportion of KD loss when maintaining a maximum learning rate can enhance model performance. (3) **Scaling law**. We find that larger student LLMs generally benefit more from pre-training distillation, and a larger teacher LLM does not necessarily guarantee better results, potentially due to the capacity gap between student and teacher LLMs (Mirzadeh et al., 2020). We further conduct PD using 500 billion tokens, and find the improvement of PD is generally consistent. (4) ***Offline* or *online***. We observe that using *online* logits for PD also yields improvement, although not as significant as *offline* logits. This suggests that one can save *online* logits on the fly during pre-training with no additional inference cost for PD on a series of smaller LLMs. In summary, we hope that our thorough exploration of the pre-training distillation design space will contribute to future practices.

2 Design Space for PD

Considering a text $\boldsymbol{x} = \{x_t\}_{t=1}^{T}$, a student LLM parameterized by θ_S, and a teacher LLM parameterized by θ_T, we formalize the objective of distillation pretraining as follows:

$$\theta_S^* = \arg\min_{\theta_S} \mathcal{L} = \arg\min_{\theta_S} [(1-\alpha)\mathcal{L}_{\text{lm}} + \alpha \mathcal{L}_{\text{kd}}] \quad (1)$$

$\mathcal{L}_{\text{lm}}$ denotes the traditional one-hot language modeling pretraining loss, which can be formalized as:

$$\mathcal{L}_{\text{lm}} = \frac{1}{T} \sum_{t=1}^{T} -\log P_{\theta_S}(x_t|\boldsymbol{x}_{<t}) \quad (2)$$

$\mathcal{L}_{\text{kd}}$ denotes the distillation loss, which can be formalized as:

$$\mathcal{L}_{\text{kd}} = \frac{1}{T} \sum_{t=1}^{T} L(P_{\theta_S}(x_t|\boldsymbol{x}_{<t}), F(P_{\theta_T}(x_t|\boldsymbol{x}_{<t}))) \quad (3)$$

L denotes the distillation loss function, such as Kullback–Leibler divergence. P_{θ_S} and P_{θ_T} represent probability of the student and the teacher LLM, respectively. F represents truncation and normalization operations conducted on the teacher LLM's logits, and τ is the temperature for normalization.

$$F(\boldsymbol{z}) = \text{softmax}(\frac{\text{Truncate}(\boldsymbol{z})}{\tau}) \quad (4)$$

Considering the key factors in Equation 1, we explore the design space of pre-training distillation in four dimensions: (1) The method F for processing the teacher LLM logits, including the truncation

method and temperature τ for normalization. (2) The choice of loss function, including the selection of distillation loss function L and the combination factor α of language modeling loss and distillation loss. (3) The scaling law of pre-training distillation, including the size of student and teacher LLMs, as well as the corpus size for pre-training the student LLM. (4) The strategy of obtaining $P_{\theta_T}(x_t|\boldsymbol{x}_{<t})$, either *offline*, i.e., the logits generated from the pre-trained teacher LLM, or *online*, i.e., the logits generated simultaneously during the teacher LLM's pre-training. In this work, we aim to conduct a systematic empirical study to investigate the impact of these four aspects on pre-training distillation and inform future practices in pre-training distillation.

3 Experiments

In this section, we conduct a preliminary experiment to introduce the basic experimental settings of pre-training distillation and validate the efficacy of pre-training distillation (§ 3.1) and empirical studies for these four main design dimensions of pre-training distillation for LLMs (§§ 3.2 to 3.5).

3.1 Preliminary Experiment

We first conduct a preliminary experiment to validate the feasibility of pre-training distillation. We use GLM-4-9B as the teacher LLM to distill of a 1.9B student LLM from scratch. To enhance training efficiency, we employ a two-stage paradigm: (1) store the teacher LLM's generated logits on the disk, (2) use these logits to train the student LLM.

Experimental Setup We first pre-train a 1.9B student LLM using pre-training distillation, namely **LLM-KD**. Specifically, we randomly sample 100 billion tokens as pre-training data. We then obtain their logits from the teacher LLM and keep the text chunk size as 4096, which is the same as the pre-training context length of the student LLM. Due to the large vocabulary size (approximately 150k items), storing the logits of the whole vocabulary using float32 requires around 58.6 PB of disk space, which is unaffordable. To reduce storage resources, we truncate the logits: we first select the top-p (Holtzman et al., 2019) logits with $p = 0.95$, and then use top-k truncation with $k = 100$, resulting in a $4,000\times$ reduced storage requirement of approximately 15 TB disk space for the 100B tokens. We re-normalize the logits with temperature $\tau = 1.0$. We use negative log-likelihood loss to conduct pre-training distillation, i.e., set $\alpha = 1$ in Equation 1 and set $L = -F(P_{\theta_T}(x_t|\boldsymbol{x}_{<t})) \log P_{\theta_S}(x_t|\boldsymbol{x}_{<t})$ in Equation 3, where F denotes our logits truncation method with a re-normalization with temperature $\tau = 1.0$. Given the limited capacity of the student LLM, its performance on some evaluation datasets, such as MMLU (Hendrycks et al., 2021) and C-Eval (Huang et al., 2024), is close to random guessing, making the results incomparable. Therefore, we conduct supervised fine-tuning (SFT; Ouyang et al., 2022) with additional 10B high-quality instruct-tuning data after pre-training. In the SFT stage of these 10B tokens, we employ only language modeling loss rather, i.e., set $\alpha = 0$ in Equation 1. We employ the same settings as in pre-training distillation, except that we only use language modeling (LM) loss for pre-training a baseline 1.9B LLM for comparison, namely **LLM-LM**. We conduct pre-training with Adam optimizer (Kingma, 2014), $2,048$ batch size, $4,096$ max sequence length, a cosine learning rate scheduler with 6×10^{-4} maximum learning rate, 6×10^{-5} minimum learning rate, and 1% warmup rate. More experimental details are placed in appendix A.1.

Evaluation Datasets We select several representative datasets to evaluate the pre-trained LLMs, including English language understanding and commonsense reasoning datasets: HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2020), PIQA (Bisk et al., 2020), MMLU (Hendrycks et al., 2021); Chinese language understanding and commonsense reasoning datasets: KBQA (Duan, 2016; Duan and Tang, 2018), C3 (Sun et al., 2020a), C-Eval (Huang et al., 2024); and math dataset: GSM8k (Cobbe et al., 2021). When conducting evaluation, the sampling temperature is set to 0. More evaluation details are shown in appendix A.1.

Experimental Results The performance of pre-trained LLM-LM and LLM-KD is presented in Table 1. We can observe that generally LLM-KD performs better than LLM-LM, though the improvement is marginal, indicating that pre-training distillation is feasible, but the current distillation configurations may not be optimal. Therefore, in the following sections (§§ 3.2 to 3.5), we will explore the design space of pre-training distillation to identify more effective configurations.

3.2 Design Dimension #1: Logits Processing

This section explores the impact of logit processing in pre-training distillation, specifically F in Equation 1, including the method for truncating logits

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
LLM-LM	53.3	54.8	72.9	28.0	3.6	54.7	25.9	8.6	37.7
LLM-KD	54.2	55.2	72.5	27.8	3.5	55.8	26.7	10.8	38.3
Δ	↑ 1.7%	↑ 0.7%	↓ 0.5%	↓ 0.5%	↓ 1.3%	↑ 1.9%	↑ 3.2%	↑ 24.6%	↑ 1.6%

Table 1: Preliminary experimental results on the evaluation datasets. Δ is relative to LLM-LM.

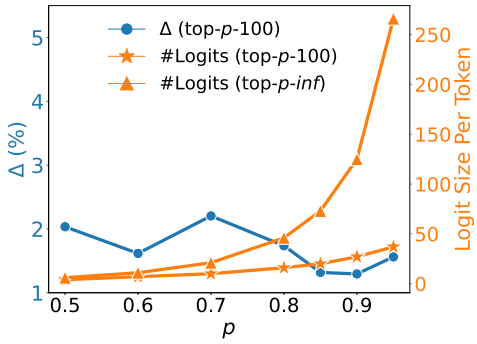

Figure 2: Relative improvements compared to LLM-LM using different p in top-p-100 logits truncation and logits sizes per token with different p. The sizes are estimated using 10 million tokens.

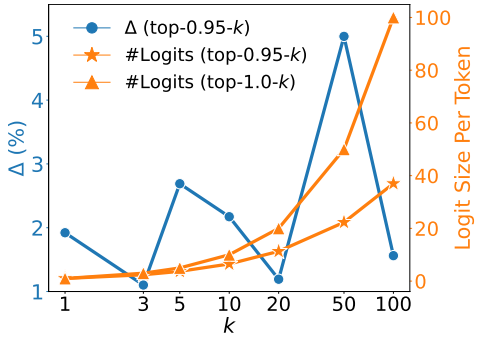

Figure 3: Relative improvements compared to LLM-LM using different k in top-0.95-k logits truncation and logits sizes per token with different k.

and the temperature τ for normalization. If not stated otherwise, all experiments adopt the same setup as the preliminary experiment, except for the processing of logits. More experimental details and results are placed in appendix A.2.

Logits Truncation As mentioned in the preliminary experiment (§ 3.1), storing the logits of the entire vocabulary requires significant disk storage space. To save resources, we design a two-stage top-p-k truncation method: truncating with top-p first, followed by top-k truncation. When the logits distribution is sharp, top-p truncation is enough; when the distribution is more uniform with long-tailed non-trivial values, top-k truncation works as a secondary truncation. Compared to vanilla top-p and top-k truncation, the top-p-k method significantly reduces storage space, as shown in Figure 2 and 3. In this section, we empirically investigate the impact of different p and k. Specifically, we set $k = 100$ to study the impact of varying p on top-p-100 truncation, and set $p = 0.95$ to analyze the effect of different k values on top-0.95-k truncation. The results are shown in Figure 2 and 3. We can observe that (1) for top-p-100 truncation, different p leads to similar improvements. A possible explanation is that in distillation pre-training, student LLM primarily captures the mass of the logits. This suggests that a smaller p can be used to further reduce storage space. (2) For top-0.95-k truncation, all values of k lead to improvements, with $k = 50$ yielding the best results. For $k = 1$, which is adopted in AFM pre-training (Gunter et al., 2024), is equivalent to using the LM loss but with labels generated from the teacher LLM and also yields an improvement. This may be due to the teacher LLM conducting implicit noise filtering in pre-training corpora. In general, pre-training distillation with different p and k values in top-p-k truncation shows improvements with limited differences, and one can adopt smaller p and k in logits truncation to save storage disk space.

τ	0.05	0.1	0.2	0.5	1.0	2.0	5.0	10.0
↑	1.6	2.1	2.5	2.7	1.6	2.5	−0.1	1.0

Table 2: Relative improvements (%) compared to LLM-LM using different τ in logits normalization.

Temperature τ Another factor is the temperature τ in logits normalization. A lower temperature sharpens the logits distribution, while a higher temperature results in a more uniform distribution. We first examine the impact of different static τ, as shown in Table 2. We can observe that lower temperatures ($\tau \leq 2.0$) lead to similar improvement, whereas at higher temperatures ($\tau \geq 5.0$), the improvement is limited. This suggests that learning from a more uniform distribution may be not efficient for student LLM. We also explore adaptive temperature, where temperature dynamically adjusts based on each sample, i.e., each token in

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average	Δ
NormKD	51.2	54.1	71.0	26.6	3.2	54.6	29.0	8.0	37.2	↓ 1.3%
WTTM	51.4	56.2	72.9	26.7	3.6	55.1	27.3	9.2	37.8	↑ 0.2%
AdaKD_{SD}	54.7	54.5	73.0	25.7	3.7	56.1	25.9	11.8	38.2	↑ 1.2%
AdaKD_{H}	54.7	57.7	73.4	25.6	3.7	57.0	27.0	10.9	38.8	↑ 2.8%

Table 3: Experimental results of LLMs pre-trained with different adaptive temperature τ methods.

pre-training distillation. We investigate two representative methods: NormKD (Chi et al., 2023) and WTTM (Zheng and YANG, 2024). NormKD applies adaptive temperature to both teacher and student logits, while WTTM applies only to the teacher logits. In this experiment, along with temperature τ, the loss calculation method is also modified. For details, refer to their original papers, and relevant hyper-parameters in loss calculation are listed in appendix A.2. We also implement a compact version of the adaptive temperature method, named AdaKD, which applies a higher temperature to smooth sharper teacher logits and a lower temperature for less sharp logits to help the student LLM focus on the most important parts (Wei and Bai, 2024). We use standard deviation and entropy to measure the sharpness of the logits, referred to as AdaKD_{SD} and AdaKD_{H}, and adaptively calculate the temperature accordingly. AdaKD_{SD} adopts the standard deviation as the temperature τ. AdaKD_{H} adopts τ_H in Equation 5.

$$\tau_H = \tau_{\text{max}} - (\tau_{\text{max}} - \tau_{\text{min}}) \times \frac{H}{H_{\text{max}}} \tag{5}$$

H denotes the entropy of each sample. More experimental details are placed in appendix A.2. The results are presented in Table 3. We can observe that AdaKD_{H} performs the best, but compared to static temperature ($\tau = 0.5$), adaptive temperature does not show significant additional improvement.

3.3 Design Dimension #2: Loss Selection

This section explores loss selection in pre-training distillation, including the types of distillation loss L and the selection of combinations with the LM loss, i.e., α in Equation 1. For all the experiments in this section, all settings remain the same as those in the preliminary experiment, except for the choice of loss. More results are placed in appendix A.3.

Distillation Loss Function L We first explore the impacts of different distillation loss functions L. Specifically, we examine three common-used types of loss function: negative log-likelihood (NLL) as used in the preliminary experiment (§ 3.1), Kullback–Leibler divergence (KLD), and mean squared error (MSE) loss. To control for variables, we omit LM loss and only use the distillation loss, setting $\alpha = 1$ in Equation 1. The experimental results are presented in Table 5. We can find that the LLMs trained with NLL and KLD loss both perform better than LLM-LM. While LLM-KLD generally outperforms LLM-NLL, the latter demonstrates superior performance on more challenging datasets, such as MMLU and C-Eval. The student LLM trained with MSE loss exhibits a significant performance decline, as observed in previous studies (Muralidharan et al., 2024). This finding contrasts with prior research in image classification (Kim et al., 2021), which finds MSE loss is the most superior choice in knowledge distillation, indicating that the pre-training distillation of LLMs involves new training dynamics and requires further investigation.

α	0.1	0.5	0.6	0.7	0.8	0.9	0.95	1.0
↑	0.1	1.5	1.4	2.9	2.0	3.6	2.5	1.6

Table 4: Relative improvements (%) compared to LLM-LM using different α in combination of $\mathcal{L}_{\text{lm}}$ and $\mathcal{L}_{\text{kd}}$.

Combination of $\mathcal{L}_{\text{lm}}$ and $\mathcal{L}_{\text{kd}}$ We examine the impact of different combinations of $\mathcal{L}_{\text{lm}}$ and $\mathcal{L}_{\text{kd}}$. We set $\mathcal{L}_{\text{kd}}$ as negative log-likelihood loss for all experiments. Specifically, we first explore the effect of different values of static α, ranging in $\{0.0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0\}$. The results are shown in Table 4, and we can observe that as α increases, the PD performance improves generally, then declines, with the best performance at $\alpha = 0.9$. This suggests that while a higher proportion of distillation loss can boost the distillation performance, an appropriate ratio (about 10%) of LM loss can further enhance pre-training distillation performance.

We further explore dynamic scheduling of α in the following ways: (1) α linearly increases from 0 to 1, namely Linear Inc, or decreases from 1 to 0, namely, Linear Dec, during pre-training. The intuition of the former is that training initially with

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average	Δ
0-α+WSD-LR	54.1	55.1	73.1	27.5	3.8	55.6	27.5	8.5	38.2	↑ 1.2%
LLM-NLL	54.2	55.2	72.5	27.8	3.5	55.8	26.7	10.8	38.3	↑ 1.6%
LLM-KLD	55.3	56.7	73.5	26.7	3.6	56.7	25.4	11.5	38.7	↑ 2.6%
LLM-MSE	44.6	55.0	69.6	25.2	2.8	52.2	25.6	3.9	34.9	↓ 7.6%
Linear Inc	53.6	55.2	73.1	25.9	3.4	56.4	28.9	8.5	38.1	↑ 1.1%
Linear Dec	53.4	56.6	72.9	29.6	3.6	56.0	30.5	11.4	39.2	↑ 4.1%
Period	52.9	55.0	72.3	28.4	3.4	55.1	27.9	9.4	38.0	↑ 0.9%
1-α+WSD-LR	56.1	57.2	73.6	27.0	3.8	58.3	29.1	11.6	39.6	↑ 5.0%
WSD-α+Cos-LR	54.0	55.4	72.7	25.1	3.7	57.6	29.4	10.6	38.6	↑ 2.3%
WSD-β+WSD-LR	53.1	55.2	73.7	27.5	3.6	55.7	25.0	11.2	38.1	↑ 1.1%
WSD-α+WSD-LR	56.4	57.7	73.6	31.8	2.6	57.6	33.8	12.5	40.7	↑ 8.0%

Table 5: Experimental results of LLMs pre-trained with different pre-training loss. Δ is relative to LLM-LM. 0-α and 1-α denote setting $\alpha = 0$ and $\alpha = 1.0$, respectively. 0-α+WSD-LR represents LLM-LM training with the WSD scheduler, which serves as a baseline. Cos-LR means a cosine learning rate scheduler. $\beta \equiv 1 - \alpha$, and WSD-β denotes applying the WSD scheduler to the proportion of LM loss.

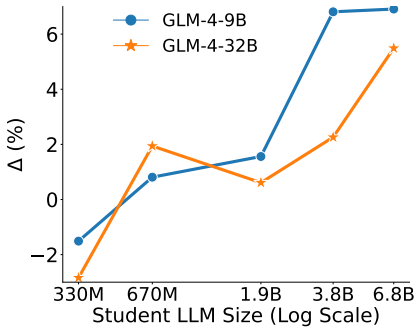

Figure 4: Relative improvements compared to LLM-LM using varying sizes of student and teacher LLMs.

LM loss may help mitigate the effects of the capacity gap with the teacher LLM; the latter is that using KD loss first may provide better optimization initialization (Yim et al., 2017). (2) α periodically varies between 0 and 0.9, namely Period, setting α to 0.9 at every fourth batch and 0 for the other batches (Kiefel and Shah, 2024). (3) We employ a nonlinear scheduler, warmup-stable-decay (WSD; Hu et al., 2024), for scheduling α, namely WSD-α. Specifically, we first linearly increase α from 0 to 1.0 during the warm up stage, then stay $\alpha = 1.0$, and finally apply cosine decay to reduce α from 1.0 to 0. We set the warmup ratio at 10% and the decay ratio at 1%. Furthermore, we employ the WSD learning rate scheduler (Hu et al., 2024), namely WSD-LR, setting its warmup and decay ratios as those of WSD-α. The intuition is that when the learning rate stays at its maximum, utilizing KD loss may enhance training efficiency. The results are shown in Table 5. We can observe that: (1) A linear decrease in α outperforms a linear increase, indicating that involving more KD loss in the early pre-training stage is more beneficial. (2) The WSD learning rate scheduler generally provides benefits, with greater gains when combined with KD loss. (3) The WSD α scheduler with the WSD learning rate scheduler yields the best performance, and the improvement of WSD β ($\beta \equiv 1 - \alpha$) scheduler with WSD-LR is limited, suggesting that using KD loss when maintaining a high learning rate effectively enhances model performance. Compared to WSD-LR with only KD loss, WSD-α performs better, indicating that a small proportion of LM loss can further enhance distillation performance.

3.4 Design Dimension #3: Scaling Law

We investigate the scaling law of pre-training distillation, including the impact of varying sizes of student and teacher LLMs, as well as the pre-training corpus size. All experimental settings are the same as the preliminary experiment, except for the sizes of LLMs and pre-training corpus. More experimental details are placed in appendix A.4.

Model Size We first investigate the performance with varying sizes of student and teacher LLMs in pre-training distillation. Specifically, we adopt teacher LLMs of 9B and 32B to distill student LLMs of 330M, 670M, 1.9B, 3.8B, and 6.8B. For each size of the student LLM, we pre-train a baseline LLM using only the LM loss, i.e., setting $\alpha = 0$ in Equation 1. The relative improvements compared to baseline LLMs are illustrated in Figure 4. We can observe that: (1) Larger student LLMs generally benefit more from pre-training distillation. (2) Distilling from a larger teacher LLM does not necessarily yield better performance. This may be due to the capacity gap between teacher and student LLMs (Mirzadeh et al., 2020; Gou et al., 2021). Increasing the student LLM size or using a

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average	Δ
LLM-Online-100B-L	30.1	53.0	62.1	24.5	0.7	40.2	25.9	2.4	29.8	↓ 20.9%
LLM-Online-100B	49.5	54.2	70.5	25.2	3.0	54.2	25.5	8.0	36.3	↓ 3.9%
LLM-Online-100B*	52.9	55.4	72.3	26.6	3.6	57.0	25.4	10.0	37.9	↑ 0.5%

Table 6: Experimental results of different LLMs pre-trained with *online* logits. Δ is relative to LLM-LM.

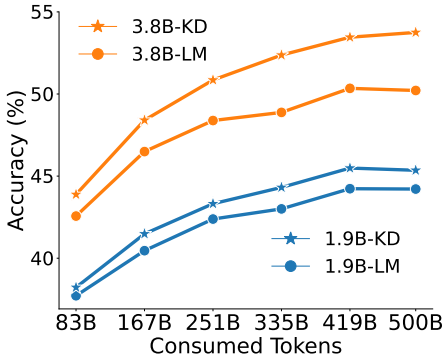

Figure 5: Experimental results of the checkpoints saved every $10,000$ step (about 83B tokens) during the pre-training of 1.9B and 3.8B LLMs on 500B tokens. The last data point is from the checkpoint saved at the end.

smaller teacher LLM can both reduce this gap and hence improve distillation performance. From a compression perspective, larger LLMs compress information more effectively and achieve better compression rates (Deletang et al., 2024), potentially making it harder for smaller LLMs to learn. Our experiments demonstrate that pre-training distillation is effective when the size of the student LLM reaches about 10% or more of the teacher LLM size, and as the proportion increases, the benefits of pre-training distillation grow until reaches the turning point. Due to computational constraints, we do not explore the turning point of performance gain to the proportion, which we leave as future work. Furthermore, scaling the student LLM to larger sizes may yield new interesting findings, such as the weak-to-strong generalization (Burns et al., 2024): using a small teacher LLM to help train a large student LLM. Due to computational constraints, we leave these explorations as future work.

Corpus Size We further investigate the impact of pre-training corpus size. Specifically, we use GLM-4-9B as the teacher LLM and distill 1.9B and 3.8B student LLMs with 500 billion tokens. We also pre-training corresponding baseline LLMs with only LM loss. We save a checkpoint every $10,000$ optimization step (about 83B tokens) and save the last checkpoint at the end of pre-training. All the other settings are consistent with the preliminary experiment. The results are illustrated in Figure 5. We can observe that: (1) Compared to student LLMs trained only with LM loss, pre-training distillation consistently yields improvements throughout the pre-training process, remaining effective with more tokens. (2) The gains from pre-training distillation increase initially during pre-training and then converge with a slight decrease, and are still significant are the end of pre-training. This suggests that pre-training distillation not only enhances training efficiency but also improves the performance upper bound of student LLMs. Due to computational limitations, we do not reach trillion-level tokens for pre-training which are used by most advanced LLMs (Team et al., 2024; Dubey et al., 2024; Gunter et al., 2024; GLM et al., 2024; Team, 2024; Liu et al., 2024). We believe that pre-training distillation is also effective using several trillion tokens and encourage future LLM development to incorporate pre-training distillation.

3.5 Design Dimension #4: *Offline* or *Online*

This section explores how logits are obtained, either *offline* or *online*. *Offline* means that logits are obtained from a pre-trained teacher LLM, which is the setting for all previous experiments. *Online* refers to storing logits generated simultaneously during the pre-training of the teacher LLM. The advantage of *online* is that it does not require additional inference from the teacher LLM if one stores the logits during teacher pre-training. Another potential advantage is that learning from *online* logits is similar to curriculum learning (Soviany et al., 2022), which may help mitigate the capacity gap and improve learning efficiency. Due to the high cost of pre-training GLM-4-9B from scratch, we preliminarily pre-train GLM-4-9B from scratch using 400 billion tokens while storing the logits for each token. We first distill two 1.9B student LLMs using the setup in § 3.1: LLM-Online-100B-L and LLM-Online-100B, which adopt the first and the last 100 billion tokens during teacher LLM's pre-training process, respectively. Experimental details are presented in appendix A.5. The results are presented in Table 6. Both LLMs yield poor performance, particularly

LLM-Online-100B-L. The reason may be that the teacher LLM is far from convergence, and hence the logits contain substantial noise. We adjust the loss calculation with $\alpha = 0.1$ and use top-0.95-50 truncation to train LLM-Online-100B*, which performs slightly better than LLM-LM, although it still underperforms LLM-KD using *offline* logits. This indicates that even logits generated by a non-converged teacher LLM can help pre-training student LLM, suggesting that using *online* logits is also effective and better practice is to utilize the logits from the later stages of the teacher LLM's pre-training. We suggest that if one aims to pre-train only an LLM, using *offline* logits of a pre-trained teacher LLM is better; if one aims to pre-train a series of LLMs of varying sizes, one can first pre-train the largest LLM while storing *online* logits, and then pre-train smaller LLMs with *online* logits.

4 Related Work

Knowledge distillation aims to transfer knowledge from a large teacher model into a smaller student model for model compression. It is first formalized by Hinton (2015), which adopts the teacher model's logits as soft targets to train the student model, which can provide richer information (Gou et al., 2021) and is also similar to label smoothing (Kim and Kim, 2017) and regulation (Müller et al., 2019; Ding et al., 2019). In this paper, we also focus on logits-based knowledge distillation. Knowledge distillation has been widely applied in in computer vision (Komodakis and Zagoruyko, 2017; Ahn et al., 2019; Wang et al., 2020b; Bergmann et al., 2020; Zhao et al., 2022; Habib et al., 2023), natural language processing (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020a; Chen et al., 2020; Taori et al., 2023; Xu et al., 2024), and speech recognition (Chebotar and Waters, 2016; Fukuda et al., 2017; Tan and Wang, 2021) domains.

Since the emergence of ChatGPT (OpenAI, 2022), knowledge distillation has become one of the most crucial techniques for enhancing large language models (LLMs). Typically, KD is applied during the post-training phase in sequence-level (Kim and Rush, 2016) to efficiently align them with humans (Xu et al., 2024), where student LLMs are trained using a teacher-forcing language modeling loss from instructions and corresponding responses generated by advanced proprietary LLMs, such as GPT-4 (OpenAI, 2023). Alpaca (Taori et al., 2023) is the first public LLM distilled from ChatGPT, providing a practical approach for improving open-source LLMs. Due to the compactness and efficacy of post-training KD, it is widely applied in developing various LLMs (Xu et al., 2023; Taori et al., 2023; Vicuna, 2023; Mitra et al., 2023; Ding et al., 2023; Sun et al., 2024; Qi et al., 2024; Cui et al., 2024), which significantly advances the development of LLMs.

For pre-training distillation of language models, there are two main categories of related work: (1) Distilling small language models in the pre-ChatGPT era (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020a; Xu et al., 2020; Sun et al., 2020b; Zhang et al., 2020; Liu et al., 2020; Hou et al., 2020). These approaches are usually based on models with only several million parameters, such as BERT (Kenton and Toutanova, 2019), and hence their training configurations may not be directly applicable for billion-level LLMs. (2) Distilling LLMs (Gu et al., 2024; Muralidharan et al., 2024; Kiefel and Shah, 2024; Turuvekere Sreenivas et al., 2024; Team et al., 2024; Gunter et al., 2024). MiniLLM (Gu et al., 2024) is trained based on a pre-trained LLM rather than from scratch. Gemma 2 (Team et al., 2024), AFM (Gunter et al., 2024), LokiLM (Kiefel and Shah, 2024), and Minitron (Turuvekere Sreenivas et al., 2024) employ pre-training distillation but provide limited details on the distillation process. While Muralidharan et al. (2024) explores the best practices for pruning and distillation of LLMs, it mainly focuses on pruning and does not systematically explore pre-training distillation. In this work, we systematically explore the design space of pre-training distillation and conduct extensive experiments to find key impact factors and better configurations. Our findings can also be applied to previous pruning and distillation work, and we hope these explorations could inform future practices in pre-training distillation.

5 Conclusion

In this paper, we systematically explore the design space of pre-training distillation, including four main impacting factors: logits processing, loss selection, scaling law, and strategies for obtaining logits, i.e., *offline* or *online*. We conduct extensive experiments to study each design dimension and identify better configurations. We also draw some interesting conclusions, such as larger student LLMs generally benefiting more from pre-training distillation while larger teacher LLMs do not guar-

antee better results. We hope our exploration will inform future practices in pre-training distillation.

Limitations

The main limitation of this work is that we do not explore the interactions between different factors in pre-training distillation, that is, the different combinations of factors. This is unaffordable, as these experiments are too resource-intensive given the complexity of factor combinations. Our controlled variable experiments have already incurred significant computational costs, which emit a significant amount of carbon dioxide and negatively impact the environment (Strubell et al., 2019). While searching the combinations of factors could identify best practices, we believe our experiments and explorations are sufficiently solid to inform future practices in pre-training distillation.

Ethical Considerations

We discuss the ethical considerations of this work: (1) Intellectual property. We strictly adhere to the copyright licenses of all the used models and datasets. (2) Intended use. Our work explores the design space of pre-training distillation, aiming to inform future practices in pre-training distillation. (3) Potential risk control. We believe the data used has been properly anonymized. As an empirical study, we do not publish additional artifacts. (4) AI assistance. We adopt ChatGPT for paraphrasing some sentences and grammar checks.

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A Experimental Details and more Results

This section introduces the experimental details and additional results. All experiments are conducted on Nvidia H800 GPUs.

A.1 Preliminary Experiment

The architecture of the 1.9B student LLM is shown in Table 7. For the SFT phase, we utilize a mixture of 10B high-quality instruction-tuning data and an additional 10B pre-training text corpus. For the instruction-tuning data, we only compute the language modeling loss for the response part. In the SFT stage, we adopt a 256 batch size, a cosine learning rate scheduler with 4×10^{-5} maximum learning rate, 4×10^{-6} minimum learning rate, and 1% warmup rate. As for evaluation, we adopt zero-shot evaluation for HellaSwag, WinoGrande, PIQA, and KBQA; 5-shot evaluation for C3 and C-Eval; 6-shot evaluation for MMLU; and 8-shot evaluation for GSM8k. We set the sampling temperature to 0.

A.2 Logits Processing

We first employ NormKD (Chi et al., 2023) and WTTM (Zheng and YANG, 2024) as the adaptive temperature calculation methods. Our implementation differs slightly from the original versions, as we use truncated logits instead of logits of the entire vocabulary. For NormKD, we set the hyper-parameter `T_norm` to 1.0 and α to 0.5 in Equation 1. For WTTM, we set the hyper-parameters γ to 0.1 and β to 1.0. For τ_H in Equation 5, H denotes the entropy of each sample, and H_{max} is the largest entropy and is estimated on 10 million tokens. We set $\tau_{\text{max}} = 2.0$, $\tau_{\text{min}} = 0.1$, and $H_{\text{max}} = 4.8$. Experimental results of § 3.2 on all evaluation datasets are presented in Table 8 and 9.

A.3 Loss Selection

For the WSD scheduler (Hu et al., 2024), we adopt a linear scheduler during the warmup stage and a cosine scheduler during the decay stage. The experimental results using different α on all the evaluation datasets are shown in Table 10.

A.4 Model Size

The architectures of different sizes of student LLMs are shown in Table 7. When pre-training 1.9B and 3.8B student LLMs on 500 billion tokens, we save a checkpoint every $10,000$ optimization step. We also save the checkpoint at the end. For each checkpoint, we conduct SFT as in the preliminary experiment before evaluation. The results on all evaluation datasets are shown in Table 11 and 12. We report the averaged performance in Figure 5.

A.5 *Offline* or *Online*

We pre-train a new 9B LLM from scratch as the teacher LLM, with a $1,728$ batch size, $4,096$ max sequence length, a cosine learning rate scheduler with 6×10^{-4} maximum learning rate, 6×10^{-5} minimum learning rate, and 1% warmup rate. Due to the high cost, we only adopt 400B tokens and store their logits simultaneously, which consumes about 180TB of disk storage space. This indicates that, since the teacher LLM has not yet converged, the logits are more uniform and contain more noise.

A.6 A Better Configuration for PD

Based on our exploration, we select a better configuration for pre-training distillation. For logits processing, we use top-0.95-50 truncation and apply a temperature of $\tau = 2.0$ for normalization. For loss selection, we adopt KLD as the distillation loss and combine it with LM loss using WSD-α and WSD-LR. The WSD hyper-parameters are the same as in § 3.3, except for the maximum value of α, which is set to 0.9. We use GLM-4-9B as the teacher LLM to distill 1.9B and 3.8B student LLMs. We adopt *offline* logits for PD. The results on all evaluation datasets are shown in Table 13. We report the averaged performance in Figure 1.

	Hidden Size	FFN Hidden Size	#Layers	#Attention Heads	#Query Groups	Tie
330M	$1,024$	$4,096$	12	16	2	True
670M	$1,024$	$4,096$	24	16	2	False
1.9B	$2,048$	$6,912$	24	16	2	False
3.8B	$3,072$	$8,192$	28	24	8	False
6.8B	$4,096$	$12,800$	28	32	8	False

Table 7: Model architectures of student LLMs of varying sizes. "#Query Groups" denotes the number of query groups in grouped-query attention (GQA, Ainslie et al., 2023). "Tie" represents whether to tie the word embeddings and output weights. All the models are trained with BFLOAT16 (Kalamkar et al., 2019) format.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
top-0.5-100	54.2	55.8	72.9	27.1	3.6	56.3	28.1	9.8	38.5
top-0.6-100	55.2	55.0	73.7	27.2	2.0	56.6	25.9	11.0	38.3
top-0.7-100	54.4	57.5	72.7	27.8	2.9	56.7	27.0	9.4	38.5
top-0.8-100	54.4	56.7	72.5	27.0	3.5	56.0	26.2	10.6	38.4
top-0.85-100	54.6	53.7	73.6	26.2	3.4	56.5	26.8	10.8	38.2
top-0.9-100	53.7	54.9	72.7	27.9	3.5	55.5	28.2	9.2	38.2
top-0.95-1	52.4	55.6	72.6	27.1	3.6	56.6	28.2	11.4	38.4
top-0.95-3	53.3	56.6	72.7	27.9	2.3	55.9	25.8	10.5	38.1
top-0.95-5	53.8	55.7	73.0	28.5	3.6	56.4	29.0	9.7	38.7
top-0.95-10	54.4	54.2	72.9	28.8	4.0	56.0	27.3	10.7	38.5
top-0.95-20	53.8	56.2	73.9	26.3	2.8	57.4	24.2	10.6	38.2
top-0.95-50	54.0	54.1	72.9	33.2	3.9	55.9	31.5	11.2	39.6
top-0.95-100	54.2	55.2	72.5	27.8	3.5	55.8	26.7	10.8	38.3

Table 8: Experimental results on all the evaluation datasets using different p and k in top-p-k truncation.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
$\tau = 0.05$	53.1	57.0	72.0	29.2	3.4	55.8	26.8	9.2	38.3
$\tau = 0.1$	52.6	54.2	72.6	28.6	2.6	56.1	30.6	10.8	38.5
$\tau = 0.2$	53.5	56.9	73.2	27.8	3.6	56.2	27.3	10.8	38.7
$\tau = 0.5$	54.7	57.0	74.2	28.2	3.9	56.1	26.0	9.8	38.7
$\tau = 1.0$	54.2	55.2	72.5	27.8	3.5	55.8	26.7	10.8	38.3
$\tau = 2.0$	54.1	56.7	73.2	27.8	3.7	56.2	27.0	10.5	38.7
$\tau = 5.0$	52.5	55.8	72.8	23.5	3.3	56.2	27.9	9.6	37.7
$\tau = 10.0$	52.1	57.1	73.0	27.3	3.3	53.9	30.2	8.0	38.1

Table 9: Experimental results on all the evaluation datasets using different τ in logits normalization.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
$\alpha = 0$	53.3	54.8	72.9	28.0	3.6	54.7	25.9	8.6	37.7
$\alpha = 0.1$	53.4	56.0	72.9	26.4	3.2	55.8	24.1	9.6	37.7
$\alpha = 0.5$	53.8	54.4	72.6	26.9	3.4	55.9	29.8	9.6	38.3
$\alpha = 0.6$	53.7	55.7	73.4	27.8	3.4	54.4	28.8	8.6	38.3
$\alpha = 0.7$	53.6	56.6	73.4	28.5	3.8	55.0	29.6	10.1	38.8
$\alpha = 0.8$	54.3	56.6	72.4	28.2	3.8	55.5	26.6	10.5	38.5
$\alpha = 0.9$	55.1	57.4	73.0	29.6	3.5	57.2	25.6	11.1	39.1
$\alpha = 0.95$	53.4	57.1	72.1	28.7	3.4	56.4	28.4	9.7	38.7
$\alpha = 1.0$	54.2	55.2	72.5	27.8	3.5	55.8	26.7	10.8	38.3

Table 10: Experimental results on all the evaluation datasets using different α in Equation 1.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
Baseline: LM Loss									
330M	37.4	54.1	67.4	24.0	2.0	47.3	26.2	2.3	32.6
670M	42.3	51.9	68.6	26.7	2.3	48.9	24.8	3.0	33.6
1.9B	53.3	54.8	72.9	28.0	3.6	54.7	25.9	8.6	37.7
3.8B	59.0	57.8	75.4	34.5	4.6	57.8	33.4	13.7	42.0
6.8B	63.0	59.9	75.5	36.7	4.6	61.8	37.1	20.9	44.9
Teacher LLM: GLM-4-9B									
330M	37.7	51.8	68.8	23.5	1.8	45.8	25.2	2.1	32.1
670M	43.4	50.9	69.4	25.7	2.4	49.4	26.2	3.1	33.8
1.9B	54.2	55.2	72.5	27.8	3.6	55.8	26.7	10.8	38.3
3.8B	61.4	60.2	75.6	39.1	5.0	61.0	39.5	17.1	44.9
6.8B	66.0	62.3	76.3	41.2	5.7	64.4	43.0	25.5	48.0
Teacher LLM: GLM-4-32B									
330M	37.1	51.5	67.4	24.2	2.0	45.2	24.5	1.4	31.6
670M	43.0	51.5	69.5	27.0	2.2	50.2	26.4	3.9	34.2
1.9B	53.7	57.9	73.4	26.2	3.4	54.6	26.3	8.0	37.9
3.8B	60.8	57.6	75.0	33.9	2.7	60.8	38.0	14.7	42.9
6.8B	66.2	62.3	76.6	41.4	5.1	63.7	41.4	22.7	47.4

Table 11: Experimental results on all the evaluation datasets of baseline LLMs trained with only LM loss and distilled LLMs using varying sizes of teacher and student LLMs.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
1.9B LLM pre-trained with LM Loss									
10,000	52.3	55.4	72.1	27.8	3.4	56.3	26.4	8.0	37.7
20,000	56.4	57.6	74.0	31.9	4.0	58.2	31.2	10.3	40.5
30,000	58.5	58.6	74.5	33.6	4.2	59.4	38.0	12.3	42.4
40,000	59.8	57.6	74.8	35.7	4.3	60.4	36.9	14.5	43.0
50,000	60.6	58.0	75.8	37.8	4.6	62.0	40.3	14.9	44.2
59,604	61.1	58.8	75.4	37.7	4.5	60.9	39.7	15.7	44.2
1.9B LLM pre-trained with KD Loss									
10,000	53.8	57.1	73.0	26.0	3.1	56.3	25.9	10.7	38.2
20,000	58.1	58.7	74.3	31.4	3.7	59.6	31.5	14.5	41.5
30,000	60.0	59.1	74.6	34.4	4.6	60.0	35.8	18.0	43.3
40,000	60.9	60.0	74.9	35.1	4.9	61.7	38.0	19.0	44.3
50,000	61.8	59.9	75.4	38.5	4.3	61.9	41.4	20.6	45.5
59,604	61.9	60.3	75.5	38.9	4.6	61.8	40.3	19.4	45.4
3.8B LLM pre-trained with LM Loss									
10,000	58.6	59.9	74.4	33.1	4.7	60.2	36.8	12.8	42.6
20,000	63.5	61.3	75.6	41.0	4.4	63.2	42.3	20.5	46.5
30,000	65.7	63.6	76.1	42.8	2.8	65.1	47.3	23.7	48.4
40,000	67.1	63.2	76.6	45.2	1.3	65.8	46.1	25.8	48.9
50,000	68.0	64.2	76.7	46.0	4.5	66.9	48.0	28.5	50.3
59,604	68.3	63.1	77.3	46.9	2.3	66.7	47.8	29.3	50.2
3.8B LLM pre-trained with KD Loss									
10,000	60.8	61.5	75.6	31.7	4.8	61.0	36.6	19.0	43.9
20,000	65.3	63.1	76.3	41.6	5.7	64.0	44.8	26.5	48.4
30,000	67.2	65.2	76.4	47.0	6.2	66.4	47.5	30.9	50.9
40,000	68.3	65.4	76.7	49.4	6.9	67.1	50.2	35.0	52.4
50,000	69.1	67.4	77.3	51.3	6.7	68.5	50.9	36.5	53.5
59,604	69.5	66.5	77.7	52.4	6.8	68.5	52.3	36.2	53.7

Table 12: Experimental results on all the evaluation datasets of different checkpoints saved every $10,000$ optimization step when pre-training the LLMs on 500 billion tokens. "59604" is the last checkpoint saved at the end.

	HellaSwag	WinoGrande	PIQA	MMLU	KBQA	C3	C-Eval	GSM8k	Average
1.9B	56.9	59.1	73.9	29.8	3.7	59.0	35.2	12.4	41.2
3.8B	62.4	61.2	76.0	38.1	5.0	62.8	38.5	21.5	45.7
6.8B	67.4	65.1	76.6	44.3	5.6	67.1	44.7	27.4	49.8

Table 13: Experimental results on all the evaluation datasets of a better pre-training distillation configuration.